MEF UNIVERSITY | COMP 204 PROGRAMMING STUDIO

projectiri

AI-Driven Spotify Curator & Cross-Platform Content Engine

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v2.0 — June 2026 · Major Update

Problem Statement

THE DATA OVERLOAD

Over 100,000 new tracks are uploaded to Spotify daily.
Human curators cannot filter them at scale.

THE MOOD GAP

Keyword search returns songs by title, not by feel.
A user saying "yorgunum ama umutlu" never gets the right music.

THE PRODUCTION GAP

Even great playlists die without covers, promotion, and cross-platform reach.

100K+

tracks added daily on Spotify

13s

average time between new uploads

Manual curation is impossible.

Project Objectives

Smart Curation

Semantic mood matching with sentence-transformers embeddings, 5 curator personas, diversity filter.

Mood Intelligence

Free-text mood → structured music profile (energy, tempo, vocals, style, what to avoid).

Cross-Platform

Spotify × MusicBrainz × YouTube cross-reference. Discover tracks missing from streaming.

Content Pipeline

AI covers (Pollinations + Nano Banana), TikTok video gen, scheduled distribution.

What We Built

Eight production modules — all live, automated, integrated.

01
Personal Playlist
Mood + name + zodiac → AI playlist

02
Auto Genre Playlists
9 genres, daily refresh, hit scoring

03
Weekly Trend Playlists
Reddit + Groq LLM → viral names

04
TikTok Video Generator
9:16 MP4 + auto-caption

05
Cover Generator
Pillow + Nano Banana + Pollinations

06
Missing Tracks Finder
Spotify × MusicBrainz × YouTube

07
YouTube Playlist Build
Missing tracks → YT playlist (OAuth)

08
Analytics Dashboard
SQLite snapshots, growth tracking

→ *All 8 modules accessible from web app + native macOS .app + Textual TUI*

→ Cron-scheduled: daily 09:00 (genres) · Monday 09:30 (trends) · daily 03:00 (analytics)

16 Python modules	**11** AI / API integrations	**3** Distinct interfaces	**30+** Public playlists

Mood Intelligence Engine

Free-text input → structured music profile → semantically-ranked tracks

01

Mood Interpreter

Groq Llama-3.3 reads the user's free text + age + zodiac.

Returns structured profile:
- primary + secondary emotion
- likely activity (study, drive, cry...)
- energy level + BPM target
- vocal style + sonic palette
- style families + what to avoid

02

Curator + Search

Profile becomes anchor signals:

- 5-6 anchor artists
- 3-5 specific tracks
- 3-5 niche sub-genres

Spotify search is run with each signal.
Persona filter (Mainstream / Underground / Throwback / Fresh) applied.

03

Semantic Ranking

sentence-transformers embeds:
- the mood description
- each candidate track

Cosine similarity score:
mood_match × 0.65 + popularity × 0.20 + recency × 0.15

Diversity cap: max 2 tracks per artist.

Result: "babamı gururlandırmak istiyorum" → uplifting indie anthems, not keyword-matched sad music.

Curator Persona System

Same mood, five distinct curatorial voices. The user picks the personality.

PERSONA	VIBE	POP	YEAR	FOCUS
DEFAULT	Balanced	Pop 30–100	2000+	Mix of mainstream + niche
MAINSTREAM	Viral Hits	Pop 60–100	2020+	TikTok / chart-toppers
UNDERGROUND	Hidden Gems	Pop 0–45	2018+	Low-popularity discoveries
THROWBACK	Aged Classics	Pop 20–100	1980–2015	Older tracks that aged well
FRESH	This Week	Pop 20–100	2025+	Newest releases only

Cross-Platform Discovery

Find tracks an artist has on YouTube but not on Spotify — demos, live cuts, deleted releases.

SPOTIFY

Live Catalog

What the artist currently has on streaming.

- Multi-query search to handle dev-mode limits
- Strict artist-name validation

● Available

MUSICBRAINZ

Complete Discography

Open music database — every release ever recorded.

- Free, no API key
- ~200–500 recordings per artist

● Reference

YOUTUBE

Popular Uploads

All videos attributed to the artist, sorted by view count.

- Duet detection (feat. / ft. / with)
- Cover/karaoke/tribute filters

● Candidate Pool

OUTPUT

JSON file per artist → optional one-click conversion to YouTube playlist (OAuth)

Content Production Pipeline

Curation alone is not enough — covers, videos, branding, distribution all automated.

VISUAL · Cover Generation

Smart Fallback Chain

1. **Pillow gradient**
   Branded 9-genre covers (instant, free)
2. **Nano Banana**
   Google Gemini AI (if billing enabled)
3. **Pollinations.ai**
   Flux model fallback (free, always works)

Symbolic prompts: name meaning + zodiac element → unique cover per playlist

MARKETING · TikTok Video Generator

1080 × 1920 · MP4 + Caption

- → Blurred cover backdrop with Ken Burns feel
- → Bold playlist title typography overlay
- → Track name + artist + Spotify CTA
- → Groq-written viral hook + hashtags
- → Ready to upload — zero manual editing

Output: /tiktok_videos/ · AirDrop → TikTok in 30 seconds

System Architecture

1

DATA LAYER

Multi-source ingestion with rate-limit handling and OAuth where needed

`Spotify API · MusicBrainz · YouTube Data API · Reddit JSON`

2

INTELLIGENCE

Mood interpretation → embedding-based ranking → persona + diversity filters

`Groq Llama-3.3 · sentence-transformers · Curator engine`

3

PRODUCTION

Cover generation, TikTok MP4 assembly, playlist creation + metadata SEO

`Pillow · Pollinations.ai · MoviePy · Spotify write ops`

4

INTERFACE & OPS

Three UIs over one backend, scheduled automation, SQLite analytics

`NiceGUI web · Textual TUI · Native macOS .app · cron`

Single backend · Three UIs · Three scheduled tasks · One source of truth

Technology Stack

APIS

Spotify Web API
Search, playlists, write, OAuth scopes

YouTube Data v3
Video search, OAuth playlist creation

MusicBrainz
Free full-discography fallback

Reddit JSON
Hot posts for trend mining

Groq API
Llama-3.3 70B — fast + free

AI / ML

sentence-transformers
all-MiniLM-L6-v2 embeddings

scikit-learn
Cosine similarity ranking

Pollinations.ai
Flux image generation (free)

Nano Banana
Gemini 2.5 Flash Image

SQLite
Embedding cache + analytics

PYTHON STACK

spotipy
Spotify client + auth manager

Pillow
Cover gradients + composition

MoviePy
9:16 MP4 assembly

NiceGUI
Modern web UI (FastAPI + Quasar)

Textual
Premium terminal TUI

Three Interfaces, One Backend

Same module set, three different ways to use it. Pick your comfort zone.

WEB APP

NiceGUI

Modern dark-themed web UI with sidebar nav, live log streaming, modal forms, analytics dashboard, and playlist browser.

- localhost:8080 — runs as a real app
- Mobile-friendly (same WiFi)
- Live subprocess output streaming

NATIVE APP

macOS .app

Real macOS application bundle with custom icon, Spotlight searchable, dock-able, double-click launch.

- Codesigned (adhoc) for Gatekeeper
- Auto health-check on launch
- Opens browser only when ready

TUI

Textual

Premium terminal UI for power users — mouse + keyboard, Spotify-green accent, instant launch.

- Keyboard-first navigation
- No browser needed
- Persona dropdown in modals

Roadmap · Delivered

All planned milestones shipped, plus three unplanned modules.

W1	W2	W3	W4	W5	W6
Setup	**Curation**	**AI Stack**	**Production**	**Discovery**	**Interfaces**
✓	✓	✓	✓	✓	✓
Spotify + OAuth + centralized auth	9-genre engine hit scoring + diversity	Mood interpreter + embeddings	Covers + TikTok video generator	MusicBrainz + YouTube cross-ref	Web app + macOS bundle + TUI

BONUS · Beyond Plan

Native macOS app bundle · Birth-date zodiac engine · Activity-aware mood interpreter

Originally planned for K-means clustering on audio features. Spotify deprecated audio-features API late 2024, so we pivoted to semantic embeddings + structured mood profiling — a stronger approach.

Team & Contributions

MEMBER	ROLE	MODULES SHIPPED
İRFAN EMRE DURMUŞ	**Backend · Pipeline · Auth**	`spotify_auth` · `oto_playlist` · `trend_playlist` · `kisisel_playlist` · `curator` · `sanatci_eksik` · `yt_playlist` · `analitik` · cron jobs
AYHAN AZRA KERVAN	**Data · AI · Mood Engine**	`mood_yorumlayici` · `burc` · curator persona system · embedding ranking · MusicBrainz integration · hit scoring
YİĞİT KARAKOZAK	**UI · Branding · QA**	`web` (NiceGUI) · `projectiri` TUI · `kapak` (Pillow + Pollinations + Nano Banana) · `tiktok_uretici` · macOS `.app` bundle

Success Criteria · Measured

We didn't just ship features. We can prove they work.

SOUNDNESS

Rate limits handled
OAuth scopes unified
Subprocess error isolation
Graceful AI fallbacks

DATA INSIGHT

Embedding similarity proven for mood matching.
MusicBrainz fills 80%+ of Spotify dev-mode gaps.

ORIGINALITY

End-to-end automation: mood text → curated tracks → AI cover → TikTok video → scheduled distribution.

DELIVERABILITY

Cron-driven daily updates
Native macOS app
.gitignore-protected secrets
Analytics for retro-tuning

100% modules shipped

16 Python modules

3 scheduled tasks

0 manual steps in daily ops

Contact: iriprojectapp@gmail.com · Version 2.0 · June 2026 · Project Group 31